EXPERIMENT 12: Run a Container from Docker Hub

1. Aim

To download a Docker image from Docker Hub and run a container using the downloaded image.

2. Procedure

1. Open a web browser and visit https://hub.docker.com/.
2. Install Docker Desktop or Docker Engine and start the Docker service.
3. Open a terminal or command prompt.
4. Verify the Docker installation using docker --version.
5. Search for the Nginx image using docker search nginx.
6. Download the official Nginx image using docker pull nginx.
7. Verify the downloaded image using docker images.
8. Run an Nginx container in detached mode and map host port 8080 to container port 80.
9. Check the running container using docker ps.
10. Open a web browser and enter http://localhost:8080.
11. Verify that the Nginx welcome page is displayed.
12. Stop the container using docker stop mynginx.
13. Remove the container if required using docker rm mynginx.

3. Program

Docker commands:

```
docker --version
docker search nginx
docker pull nginx
docker images
docker run -d --name mynginx -p 8080:80 nginx
docker ps

# Open http://localhost:8080 in a browser
docker stop mynginx
docker rm mynginx
```

4. Output of Program

```
Sample terminal output:

$ docker run -d --name mynginx -p 8080:80 nginx
<container-id>

$ docker ps
CONTAINER ID   IMAGE   STATUS
xxxxxxxx       nginx   Up

Browser: Welcome to nginx!
```

5. Result

Thus, an Nginx image was downloaded from Docker Hub, and a container was created and executed successfully.

EXPERIMENT 12: Run a Container from Docker Hub

1. Aim

To download an application image from Docker Hub, create and execute a Docker container using the downloaded image and access the running application through a web browser.

2. Procedure

1. Start the computer and open a web browser. 2. Navigate to https://hub.docker.com/ to access Docker Hub, which is a public registry for Docker images. 3. Install Docker Desktop or Docker Engine and start the Docker service. 4. Open a terminal or command prompt and verify the installation using docker --version. 5. Search for the Nginx image using the command docker search nginx. 6. Identify the official Nginx image from the search results. 7. Download the image using docker pull nginx. 8. Verify that the image has been downloaded successfully using docker images. 9. Create a container using the downloaded Nginx image. 10. Execute the container in detached mode using docker run -d --name mynginx -p 8080:80 nginx. 11. The port mapping connects port 8080 on the host computer to port 80 inside the container. 12. Verify that the container is running using docker ps. 13. Check the container status and confirm that it is in the running state. 14. Open a web browser on the host computer. 15. Enter the address http://localhost:8080 in the browser's address bar. 16. Verify that the default Nginx welcome page is displayed. 17. This confirms that the Nginx web server is running successfully inside the Docker container. 18. Stop the container using docker stop mynginx. 19. If required, remove the stopped container using docker rm mynginx. 20. Verify the final container status using docker ps -a.

4. Output of Program

Docker Container Creation: $ docker run -d --name mynginx -p 8080:80 nginx Container Verification: $ docker ps CONTAINER ID IMAGE STATUS xxxxxxxx nginx Up Browser Output: Welcome to nginx! If you see this page, the nginx web server is successfully installed and working. Explanation: Nginx is a web server that can serve web pages and handle HTTP requests. Docker runs Nginx inside an isolated container. Port mapping allows the host computer to access the web server running inside the container.

5. Result

Thus, the official Nginx image was successfully downloaded from Docker Hub. A Docker container was created and executed in detached mode, and the Nginx welcome page was accessed successfully through the host computer's web browser.